

Technical Brief

Title: Overcoming Computational Bottlenecks in Real-Time Obstacle Avoidance for Serial-Chain Manipulators

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Purpose

This brief distills the key computational and control architecture challenges that I encountered during my master’s research and outlines a technical path to solve them, culminating in an industry-grade implementation.

Abstract: Obstacle avoidance in dynamic environments poses significant computational challenges, particularly for serial chained manipulators. To address these challenges, this technical brief outlines a hierarchical control framework that integrates a high-level planner with a low-level trajectory tracker. The high-level controller operates at a lower frequency, dynamically replanning paths in response to the presence and motion of obstacles. This adaptive replanning mechanism is designed to preemptively avoid potential local minima and ensure that the overall trajectory remains feasible despite rapidly changing environmental conditions. Meanwhile, the low-level controller functions at a higher frequency, ensuring precise trajectory tracking while concurrently handling unforeseen uncertainties and abrupt obstacle movements. Together, this dual-layered strategy not only mitigates the computational burden by segregating tasks based on their dynamic requirements but also significantly enhances the robustness and reliability of obstacle avoidance for serial chained manipulators operating in complex and uncertain settings.

1 Introduction

The insights and challenges presented here build upon the research conducted during my Master’s thesis [9]. Several challenges were identified during the thesis, primarily related to computational inefficiency and communication constraints. One of the key issues was the strict timing requirement—completing computations within 0.3 ms—due to the control loop operating at a frequency of 1 kHz. This constraint became particularly problematic in scenarios involving a high number of obstacles, often resulting in packet loss and degraded performance.

A further challenge involved local minima during obstacle avoidance, particularly in scenarios involving parallel surfaces. A notable example was the “parallel face problem,” where the box-shaped gripper approached a box-shaped obstacle with parallel faces. This configuration caused the Franka robot to halt and fail to proceed along the desired trajectory.

The following section outlines potential methodologies and approaches identified through recent research that aim to address these challenges, providing a roadmap for future developments in this domain.

2 Related Work and Proposed Approaches

To address computational complexity and local minima challenges in autonomous navigation, a two-layered hierarchical controller is proposed. As illustrated in Figure 1, the high-level controller performs adaptive path replanning at a lower frequency, accommodating slower update rates of environmental obstacle information. In contrast, the low-level controller operates at a higher frequency to ensure precise real-time path tracking and to handle uncertainties, such as abrupt obstacle movements that may require reflexive maneuvers. Model predictive control (MPC) and reinforcement learning (RL) are grounded in dynamic programming principles, as detailed in [15]. These methods can be used at the high level for online and adaptive planning based on environmental data.

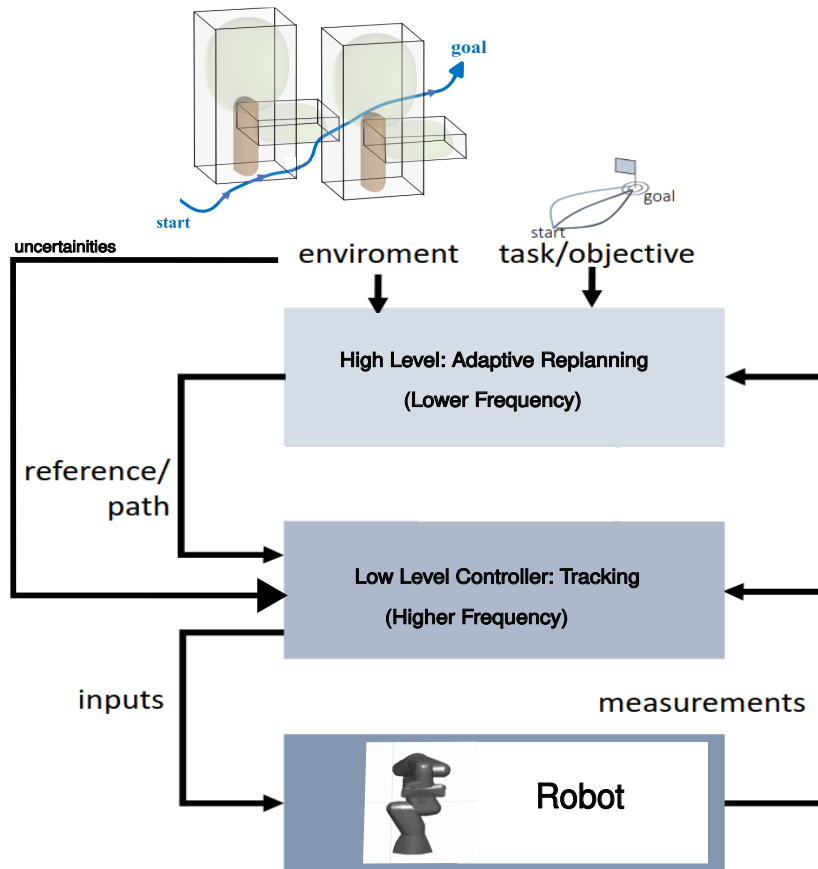


Figure 1: Hierarchical Controller Framework for Future Work, partly copied from [10]

Dynamic Programming (DP) is a fundamental technique for solving sequential decision-

making problems, especially optimal control. In a finite-horizon Markov Decision Process (MDP), the goal is to find a policy π that minimizes the cumulative cost:

$$J^\pi(x_0) = \mathbb{E} \left[\sum_{k=0}^T \ell(x_k, u_k) \right]$$

with dynamics $x_{k+1} = f(x_k, u_k)$. The value function $J_k^*(x)$ satisfies the Bellman recursion:

$$J_k^*(x) = \min_{u \in \mathcal{U}} \mathbb{E} [\ell(x, u) + J_{k+1}^*(x') \mid x' \sim P(\cdot \mid x, u)]$$

For infinite horizon problems with discount $\gamma \in (0, 1)$, the value iteration becomes:

$$J_{i+1}(x) = \min_{u \in \mathcal{U}} \mathbb{E} [\ell(x, u) + \gamma J_i(x')]$$

This iterative process converges to the optimal value function $J^*(x)$.

Model Predictive Control (MPC) is an online control strategy that solves a finite-horizon optimal control problem at each time step using the current state x_k . It computes a control sequence $u_k^*, \dots, u_{k+N-1}^*$ over a prediction horizon N , but only applies u_k^* . The optimization problem at each time step is:

$$\min_{u_0, \dots, u_{N-1}} \sum_{i=0}^{N-1} \ell(x_i, u_i) + \ell_f(x_N), \quad \text{s.t. } x_{i+1} = f(x_i, u_i)$$

MPC uses the receding horizon principle, updating the optimization as new states become available. It supports constraints and multivariable systems, but can be computationally demanding.

Reinforcement Learning (RL) learns optimal policies from interaction with an unknown environment. The stochastic dynamics are typically modeled as $x_{k+1} \sim P_X(\cdot \mid x_k, u_k)$, and the objective is to minimize the expected discounted cost:

$$J^\pi(x) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k \ell(x_k, \pi(x_k)) \right]$$

Key functions include the action-value function:

$$Q^\pi(x, u) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k \ell(x_k, u_k) \mid x_0 = x, u_0 = u \right]$$

An optimal policy can be derived as:

$$\pi^*(x) \in \arg \min_{u \in \mathcal{U}} \mathbb{E} [\ell(x, u) + \gamma J^*(x')]$$

AlAttar et al. [1] introduce a kinematic-model-free predictive controller (KMF-PC) that, without any prior robot model, learns local linear kinematic mappings online from brief exploratory actuations and embeds them into a finite-horizon MPC to achieve end-effector reaching alongside circular-obstacle avoidance. Building on hierarchical MPC for multi-rate operation, Kögel et al. [10, 4] decompose planning and control into a slow, mixed-integer linear program (MILP)-based planner—capable of selecting among “contracts” (mode-dependent error tubes) to trade off speed

and precision—and a fast cyclic-horizon tube-MPC tracker that ensures each mode’s tracking bound under bounded disturbances, thus reducing conservatism while guaranteeing recursive feasibility and safety [17, 16].

For robotic manipulators, Meza et al. [14] employ a similar two-layer hierarchy, pairing high-level path planning with a low-level inverse-kinematics linear-quadratic Model Predictive Control (LQ-MPC) for real-time obstacle avoidance using sensor-derived convex safe regions. Zhu et al. [18] further integrate artificial potential fields within a cascaded nonlinear MPC framework, where a high-level MPC generates smooth artificial potential fields (APF)-based paths and a low-level MPC enforces precise tracking and singularity-free, collision-free motion. To address high-frequency torque control, Haffemayer et al. [7] implement a hard-constraint, signed-distance MPC on a torque-controlled Panda manipulator, while Dantec et al. [5] derive a Riccati-based state-feedback approximation of the MPC law to bridge slow planning and fast actuator dynamics.

Finally, several studies demonstrate the promise of Reinforcement Learning (RL) for high-level planning in dynamic environments: Baselizadeh et al. [3] combine Nonlinear Model Predictive Control (NMPC) with RL to parameterize online cost and constraints for adaptive trajectory generation; Elguea-Aguinaco et al. [6] survey RL methods in manipulator motion planning; Almazrouei et al. [2] and Liu et al. [12] propose RL-based dynamic obstacle avoidance and deep RL path planning, respectively; Li et al. [11] integrate RL with artificial potential fields to escape local minima; and Huang et al. [8] leverage manipulator redundancy in an RL-driven obstacle-avoidance planner.

3 Implementation

Further analysis suggests that reinforcement learning (RL) is not well suited for this control architecture. RL typically struggles with continuous, high-dimensional state spaces, requires substantial training data, and performs unreliably in unfamiliar scenarios—making it impractical within the scope and timeframe of this project. Instead, this brief proposes a model predictive control (MPC)-based approach that builds on the methods developed during my Master’s thesis and supports efficient, real-time control in complex, dynamic environments.

Advantages of MPC over RL:

- Allows explicit encoding of constraints and obstacle avoidance logic
- More predictable behavior in unseen or uncertain environments
- Easier to analyze and integrate with current work

The new architecture comprises two key layers:

- **High-Level Planner (10 Hz):** Responsible for computing trajectories between start position A and end position B , while considering both static and dynamic obstacles. The planner will primarily use a linear model predictive control (MPC) scheme with a quadratic cost function. Alternatively, sampling-based methods such as Rapidly-exploring Random Trees (RRT) [13] may be employed in scenarios where offline pre-computation is practical.

- **Low-Level Tracking Controller (100 Hz):** This layer tracks the high-level trajectory and performs reactive, reflex-level obstacle avoidance under uncertainty. The controller builds directly on prior work and uses a linear tracking model over a finite horizon:

$$x(k+1) = f(x(k), u(k)) = x(k) + \Delta t \cdot J \cdot u(k) + K \cdot (x_d - x(k))$$

where u is the control input (e.g., joint velocity), and x_d is the desired state. The cost function is quadratic in u and will be minimized using dynamic programming across a receding horizon. The model predictive control (MPC) problem can be formulated as the following constrained optimization problem:

$$\begin{aligned} \min_{u_0, \dots, u_{N-1}} \quad & \sum_{i=0}^{N-1} \frac{1}{2} u_i^\top Q_i u_i \\ \text{subject to} \quad & x_{i+1} = f(x_i, u_i), \quad \forall i = 0, \dots, N-1 \\ & u_{\min} \leq u_i \leq u_{\max} \end{aligned}$$

Here, u_i is the control input, x_i is the system state, Q_i is a positive semi-definite cost matrix, and $f(x_i, u_i)$ denotes the system dynamics. The control inputs are subject to box constraints defined by $u_{\min}$ and $u_{\max}$.

This two-layer structure balances computational efficiency with system responsiveness. The high-level planner ensures global path feasibility and adapts to changes in the environment in real time, while the low-level MPC controller provides fast, local trajectory adjustments for robust tracking and reflex obstacle avoidance under uncertainty. Finally, a torque controller operating at 1 kHz will transmit torque commands to the Franka Panda manipulator, ensuring minimal communication delays and enabling smooth, responsive execution.

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